Weak-Signal Interpretation for Robotics Anomaly Sensing

A Stratified, Governance-Aware Architecture for Embodied Anomaly Handling

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Abstract

Robotic systems often encounter weak, noisy, partial, and locally ambiguous signals before a clearer anomaly becomes legible. A drive system may show slight current asymmetry, a manipulator may exhibit repeated micro-slip under otherwise plausible contact, or an environmental obstruction may appear inconsistently across perceptual routes. In such cases, the architectural problem is not only detection, but promotion control: how weak findings should influence attention before they are allowed to become stronger operational consequence. This paper argues that robotics anomaly sensing is a suitable application domain for a stratified weak-signal architecture that separates bounded observation, provisional interpretation, short-horizon attentional escalation, and governed promotion. The contribution is narrow and operational. The paper defines a minimal transfer contract for embodied anomaly handling: a light state ladder, a bounded event object, and illustrative promotion logic showing how weak signals may influence inspection before they justify stronger commitment. It does not propose a full robotics stack, semantic SLAM system, safety-certified controller, or universal diagnosis engine. Its narrower claim is that embodied systems need a disciplined place for weak anomaly signals to matter before they harden into motion change, maintenance significance, or ignored noise.

1. Introduction

Robotics anomalies rarely begin as fully resolved facts. More often they first appear as faint departures from a local corridor of expectation: a slight traction asymmetry, repeated path-correction drift, a force/torque mismatch during contact, a small thermal irregularity, an intermittent acoustic change, or disagreement across perceptual routes. Any one of these may be benign. Any one of them may also be the first visible sign that cheap continuity is no longer fully trustworthy.

That creates an architectural problem. If the robot ignores weak but structured departures because they do not yet justify confidence, it loses useful early warning. If it commits too early, it converts provisional significance into brittle state, unnecessary intervention, or unsafe overreaction. The relevant question is therefore not only whether an anomaly exists, but how a robotic system should behave when evidence is incomplete, weakly convergent, and still locally ambiguous.

This paper takes a narrow position. Robotics anomaly sensing is a suitable application domain for an architecture that separates bounded observation, provisional interpretation, attention

escalation, and governed consequence. The point is not robotic omniscience or a universal anomaly theory. The point is disciplined escalation under uncertainty in embodied systems operating under noisy, partial signals.

The contribution is architectural rather than algorithmic. This paper does not prescribe a specific robot platform, anomaly detector, control loop, or benchmark. It argues something smaller and more portable: embodied systems need a disciplined middle layer in which weak anomaly signals can raise attention before they are allowed to become stronger operational consequence.

2. Why Robotics Needs a Middle State

A robotic system lives inside contact, motion, perception, timing, and local physical constraint. That matters because many anomalies first appear not as clear failures, but as partial disturbances distributed across several bounded channels.

A drive system may show a modest current increase while one wheel path becomes slightly less smooth. A manipulator may maintain gross task success while contact becomes less stable at the micro level. A navigation system may receive inconsistent evidence about whether a corridor is open, obstructed, or merely perceptually degraded. In each case, the robot may have enough structure to justify heightened attention, but not enough to justify a durable conclusion.

Without an explicit middle state, systems tend to collapse in one of two ways. In the first, weak anomalies are treated as ordinary variance until they become obvious, by which point the value of early handling has been reduced. In the second, weak irregularities are promoted too quickly into hard alarms, degraded behavior, or false certainty, producing brittleness and overreaction. A better architecture preserves provisional significance without premature commitment.

The key point is simple: robotic systems need somewhere for weak anomaly signals to live before they harden into motion changes, maintenance significance, or ignored noise.

3. Core Claim and Design Principle

The central claim of this paper is simple:

Robotics anomaly sensing is a suitable application domain for a weak-signal architecture in which bounded findings raise attention first, candidate meanings remain provisional, and stronger consequence occurs only through persistence, convergence, and governance.

This framing does not reject sensing, estimation, control, or conventional anomaly detection. It reframes their early-stage role. In many robotic settings, the first useful evidence is too partial to deserve strong commitment but too structured to deserve indifference.

The governing design principle is equally simple:

Weak robotic anomalies should influence attention before they justify stronger operational consequence.

That means a slight wheel-drag signature, a grasp instability pattern, or a perception disagreement can widen attention, tighten inspection, or trigger bounded re-check behavior without immediately becoming durable fault state, strong environmental belief, or high-impact action.

4. Stratified Architecture Mapping

The proposed architecture separates robotics anomaly handling into four stages.

4.1 Bounded Observation

At the first layer, the system emits narrow findings only. These findings remain local to their sensing or control pathway. They may include unusual vibration, slight thermal rise, force/torque mismatch, unexpected contact behavior, wheel-current asymmetry, path-correction drift, pose instability, perception inconsistency, or disagreement across sensor routes.

At this stage, the system is not yet claiming hardware fault, hazard, or environmental truth. It is preserving local structure in a bounded form so that downstream layers can compare, combine, and challenge it.

4.2 Provisional Interpretation

At the second layer, the robot forms candidate local meanings from one or more bounded findings. These are not final truths. They are admissible interpretations of a partial pattern.

Examples include possible traction loss, possible contact instability, possible local hardware stress, possible wear precursor, possible obstacle or terrain change, possible sensor fault, possible semantic mismatch, or unresolved anomaly needing more evidence.

The key requirement is that ambiguity remain visible. The system does not become more trustworthy by pretending uncertainty has vanished. It becomes more trustworthy by representing candidate meanings in a way that remains revisable.

4.3 Short-Horizon Attention / Reflex Salience

At the third layer, weak but informative deviation influences short-horizon attention. This is the stage at which the robot may tighten an observation window, request a bounded re-check maneuver, increase local sampling, slow commitment, or seek corroboration from another channel.

The important constraint is that this layer remains temporary and non-persistent. It shapes what the system inspects next, but it does not by itself write durable memory, mutate long-horizon baselines, or authorize irreversible consequence.

4.4 Governed Consequence

Only after persistence, convergence, or stronger support builds should the architecture permit stronger consequence. At this stage, the system may allow operator review, bounded motion restriction, temporary task downgrade, sensor-stream quarantine, or maintenance-relevant logging.

The point is not that every anomaly must escalate. The point is that escalation should be explicit, inspectable, and predicate-bound rather than silently triggered by local salience alone.

A compact pipeline expresses the structure clearly:

local finding → candidate meaning → bounded attention shift → governed consequence

5. Minimal Operational Contract

This section keeps the transfer concrete without turning the paper into a full robotics platform specification.

5.1 Light State Ladder

A minimal state ladder is enough to distinguish weak-signal attention from stronger consequence:

- **R0 — Normal:** within the expected corridor for the current task or operating mode.
- **R1 — Noticed:** a weak local deviation is recorded as a bounded finding.
- **R2 — Watching:** the deviation persists across a short horizon or recurs under a controlled re-check.
- **R3 — Corroborated:** a second supporting channel confirms the concern, or a repeated maneuver reproduces the pattern.
- **R4 — Escalated:** a governed review or bounded behavior-restriction boundary has been crossed.
- **S0 — Resolved:** the deviation returns to corridor without stronger support.
- **Q0 — Quarantined:** evidence favors sensor artifact, route mismatch, or instrumentation fault rather than underlying operational anomaly.

The key boundary is simple: R1–R3 are attention and evidence states; R4 belongs to governed consequence.

5.2 Minimal Event Object

A weak-signal architecture needs a small event object that preserves structure without forcing immediate hard semantics. A minimal event object should carry:

- event identity and time bounds
- robot or subsystem identity

- task context or operating regime
- source channel or channels
- observed values and local deltas
- current ladder state
- candidate meanings
- short-horizon actions taken
- supporting channels or corroborators
- closure reason if resolved or quarantined

An illustrative event record might look as follows:

```
{
  "event_id": "evt_nav_0142",
  "robot_id": "mr_unit_03",
  "task_mode": "corridor_turn",
  "timestamp_start": "2026-04-18T10:42:11Z",
  "timestamp_end": "2026-04-18T10:42:21Z",
  "scope": "mobility/front_left_drive",
  "signals": {
    "wheel_current_delta_pct": 11.8,
    "yaw_correction_error_deg": 2.4,
    "vibration_rms_delta_pct": 9.1
  },
  "state": "R2",
  "candidate_meanings": [
    "possible traction loss",
    "possible wheel drag",
    "possible surface irregularity"
  ],
  "short_horizon_actions": [
    "tighten_sampling_window",
    "reduce_turn_speed_10pct",
    "request_recheck_maneuver"
  ],
  "supporting_channels": [
    "drive_current",
    "imu",
    "base_vibration"
  ],
  "closure_reason": null
}
```

The point is not to freeze a universal schema. The point is to preserve enough structure that weak deviations remain reviewable, comparable, and governable.

5.3 Illustrative Promotion Logic

The transition logic should remain explicit even in lightweight systems:

- **R0 → R1:** a monitored signal exceeds a scoped weak-deviation threshold and data-quality checks pass.
- **R1 → R2:** the same deviation persists for a bounded horizon or recurs under a controlled retry.
- **R2 → R3:** at least one independent corroborator supports the candidate, or a repeated maneuver reproduces the pattern.
- **R3 → R4:** governed promotion is allowed if severity remains above the scoped threshold and the rationale packet is complete.
- **R1/R2/R3 → S0:** the anomaly dissipates and no corroboration arrives.
- **Any → Q0:** evidence favors sensor artifact, mapping inconsistency, or route-specific perceptual conflict.

This logic is intentionally illustrative rather than normative. The bridge claim does not depend on a single universal rule set. It depends on keeping the promotion boundary explicit.

6. Worked Embodied Example: Wheel-Drag / Traction Anomaly

Consider a mobile robot entering a left turn in a corridor. Over a 10-second horizon, three weak departures appear together: a 12% rise in front-left wheel current, a 2–3 degree increase in corrective yaw error, and a 9% rise in local vibration RMS through the drive base.

No single signal justifies a hardware-fault declaration. The system therefore records a bounded event at R1 rather than jumping directly to a durable failure claim. Candidate meanings include possible traction loss, possible wheel drag, and possible surface irregularity.

When the same pattern persists through the remainder of the turn, the event advances to R2. At that point, the correct response is not durable fault promotion, but short-horizon shaping: narrow the sampling window, reduce turn speed modestly, and request a short re-check maneuver. These actions remain temporary and non-persistent.

If the re-check reproduces the asymmetry and a second channel confirms the pattern, the case may advance to R3 and then to governed review at R4. If the pattern disappears on re-check, the event resolves or is quarantined as surface-specific or sensor-specific noise.

The value of the architecture here is practical: the robot does not ignore the weak pattern, and it does not overread it. It gives uncertainty somewhere useful to live while the system gathers a little more evidence.

7. Two Short Supporting Examples

7.1 Grasp Instability in a Manipulator

A manipulator attempts a pick and nominally succeeds, but the episode contains three weak disturbances: force/torque mismatch at contact, repeated micro-slip during lift onset, and visual pose disagreement between expected and observed object orientation.

This is not yet a settled grasp failure, nor is it safe to pretend nothing happened. The bounded findings support several candidate meanings: possible surface slipperiness, possible grasp instability, possible object-geometry mismatch, possible contact-model error, or unresolved anomaly requiring a cautious retry. A short-horizon response may justify slower lift commitment, additional sensing during hold, or a bounded re-grasp procedure. Stronger consequence should require persistence or corroboration.

7.2 Ambiguous Environmental Obstruction Under Route Disagreement

A mobile robot approaches a corridor. The depth map suggests a partial obstruction. A bumper or tactile surface indicates weak contact. A semantic map or prior route representation still indicates the corridor should be open. The issue may reflect a real new obstacle, a reflective or deformable surface, route-specific sensor fault, or semantic mismatch across perceptual channels.

Under this architecture, the system preserves the disagreement explicitly. Short-horizon attention rises and commitment slows, but route denial or durable map change requires stronger support. The architectural point is narrow: perceptual route and operational meaning should not be blindly collapsed when anomaly evidence conflicts across channels.

8. Limits and Non-Goals

This paper is intentionally narrow.

It is not a full robotics stack.

It is not a semantic SLAM paper.

It is not a universal diagnosis engine.

It is not a safety-certified controller architecture.

It is not a replacement for controller-layer fault protection, hard interlocks, or domain-specific safety logic.

It should not be used as a standalone basis for real-time safety-critical shutdown decisions without separate certified control and protection layers.

It does not claim that weak signals will improve outcomes without domain tuning, scoped thresholds, and validation on the relevant robot, task, and environment.

Its contribution is architectural and bounded: a disciplined way to let weak embodied anomalies influence attention before they become stronger commitment.

9. Conclusion

Robotics anomaly sensing is a suitable domain for a weak-signal architecture because embodied systems routinely encounter structured deviations that are too informative to ignore and too incomplete to deserve immediate certainty. A slight traction asymmetry, grasp instability pattern, or cross-route perceptual disagreement may justify heightened attention before it justifies stronger operational consequence.

The practical value is narrow but important. A robotic system benefits from a bounded middle layer in which local anomalies may be noticed, interpreted provisionally, inspected through short-horizon salience, and promoted only through explicit support. That is the transfer claim this paper aims to make.